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Comparative effectiveness of nonpharmacological therapies for adult myogenic temporomandibular disorders: a network meta-analysis of randomized trials

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Abstract

Background Nonpharmacological therapies are widely used for adult myogenic temporomandibular disorders (TMD), but their comparative effectiveness for pain and mandibular function remains uncertain. This study compared multiple interventions using a frequentist random-effects network meta-analysis.

Methods MEDLINE (via PubMed), Embase, Cochrane Central Register of Controlled Trials (CENTRAL), Web of Science, and Google Scholar were searched from inception through October 31, 2025, without language restrictions. Randomized controlled trials enrolling adults (≥ 18 years) with myogenic TMD were included if they compared at least one predefined nonpharmacological modality (including photobiomodulation therapy [PBMT], manual therapy [MT], exercise therapy [EX], occlusal splint therapy [OST], acupuncture [ACU], electrotherapy [ELEC], combined therapy [COMB], cognitive behavioral therapy [CBT], central neuromodulation, or extracorporeal shock wave therapy) with sham or placebo control, usual care or no-treatment control, or another predefined active nonpharmacological modality. Primary outcomes were pain and maximum mouth opening, and outcome data were extracted at the short-term assessment closest to completion of the intervention. Effects were summarized as standardized mean differences (SMDs) with 95% confidence intervals (CIs) using a frequentist random-effects model implemented in Stata. Treatments were ranked using the surface under the cumulative ranking curve (SUCRA), which provides a probabilistic ranking rather than a direct measure of treatment superiority. Risk of bias was assessed using the Cochrane Risk of Bias 2.0 tool (RoB 2.0).

Results Forty-one trials (2021 participants) were included. For pain, PBMT ranked highest (SUCRA, 88.9%) and was superior to CBT (SMD, − 2.00; 95% CI, − 2.99 to − 1.00), ELEC (SMD, − 1.64; 95% CI, − 2.55 to − 0.73), conventional care (SMD, − 1.49; 95% CI, − 2.07 to − 0.91), OST (SMD, − 1.43; 95% CI, − 2.29 to − 0.56), and ACU (SMD, − 1.07; 95% CI, − 1.93 to − 0.21). MT (SUCRA, 79.9%) and COMB (SUCRA, 72.5%) also showed significant analgesic advantages versus several comparators. For maximum mouth opening (31 trials; 1463 participants), MT ranked highest (SUCRA, 92.9%) and improved opening more than conventional care (SMD, 2.79; 95% CI, 1.33 to 4.26), PBMT (SMD, 2.23; 95% CI, 0.19 to

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4.28), ACU (SMD, 2.01; 95% CI, 0.12 to 3.90), and EX (SMD, 1.85; 95% CI, 0.05 to 3.64). Inconsistency was detected for pain in the ELEC versus MT comparison; no significant inconsistency was observed for maximum mouth opening.

Conclusion Among adults with myogenic TMD, PBMT showed the most favorable short-term comparative profile for pain relief, whereas MT ranked highest for improving maximum mouth opening. These findings support targeted selection of specific nonpharmacological modalities, prioritizing PBMT when pain reduction predominates and MT when restoration of mandibular mobility is the primary goal.

Trial registration PROSPERO (CRD420261278458)

Keywords Myogenic temporomandibular disorders, Nonpharmacological treatment, Pain, Maximum mouth opening, Network meta-analysis, Randomized controlled trials

Introduction

Myogenic temporomandibular disorders (TMD) are among the most common causes of persistent, nonodontogenic orofacial pain in adults and are frequently accompanied by functional limitation, including restricted maximum mouth opening and pain during jaw movement [1]. Population based estimates suggest that clinically relevant TMD affects approximately 5% to 12% of individuals, with a higher frequency in women than in men [2]. In the United States, national estimates indicate that 5% to 10% of the population has some form of TMD and that more than 10 million Americans may be affected, with a female predominance that approaches a twofold difference [3]. Within clinical TMD populations, myofascial pain related diagnoses are reported as the most frequent subtype, accounting for nearly half of diagnoses in some cohorts, underscoring the centrality of muscle mediated pain mechanisms in routine practice. The consequences extend beyond pain intensity alone, as myogenic TMD is associated with impaired chewing, sleep disruption, psychological distress, and reduced quality of life, thereby contributing to repeated health care utilization and substantial socioeconomic burden [4, 5].

Pharmacologic management, including nonsteroidal antiinflammatory drugs, muscle relaxants, and adjuvant analgesics, is commonly used to achieve short term symptom control, particularly during acute exacerbations. However, benefits are often modest and time limited, and long-term use can be constrained by adverse events, contraindications, and concerns regarding dependence or medication overuse [6]. These limitations have reinforced the role of conservative, nonpharmacological management as a cornerstone of care. In contemporary practice, commonly used options include structured exercise-based rehabilitation, manual techniques, occlusal appliance therapy, and selected physical modalities, often delivered within multimodal programs that seek to address peripheral nociceptive drivers, motor dysfunction, and psychosocial contributors [7–9].

Despite widespread use, the comparative effectiveness of nonpharmacological therapies for adult myogenic TMD remains uncertain. The evidence base is dominated by randomized controlled trials (RCTs) that are frequently small, single center, and heterogeneous with respect to eligibility criteria, intervention dosing and technique, outcome instruments, and follow up duration [10, 11]. Many trials compare an active intervention only with conventional care or placebo rather than with another active treatment, limiting the clinical interpretability of pairwise estimates for real world decision making. Additional gaps include variable blinding, incomplete reporting of adherence and cointerventions, and inconsistent selection of pain scales and functional outcomes, all of which may increase imprecision and constrain external validity [12, 13]. Conventional pairwise meta-analyses have summarized selected interventions, but they typically evaluate only a subset of available options and cannot integrate indirect evidence across a network when head-to-head trials are sparse [14]. As a result, clinicians are often left with fragmented evidence and an inability to generate coherent comparative rankings across multiple competing nonpharmacological strategies.

The present review addressed the following research question: among adults with myogenic TMD, which nonpharmacological therapies are most effective for reducing pain and improving maximum mouth opening? This question is clinically relevant because treatment selection in routine practice often depends on whether pain relief or functional improvement is the primary goal, yet direct comparative evidence remains limited. In this systematic review and frequentist random-effects network meta-analysis (NMA), we compared multiple nonpharmacological therapies for adult myogenic TMD with respect to pain and maximum mouth opening. By integrating direct and indirect evidence and providing treatment rankings using the surface under the cumulative ranking curve (SUCRA), the findings are intended to support evidence-based selection of conservative therapies and to identify priorities for future comparative trials.

Methods

This systematic review and network meta-analysis adhered to the guidelines outlined in the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement [15] (Supplementary File 1) and was prospectively registered with the International Prospective Register of Systematic Reviews (PROSPERO; CRD420261278458). The review question was as follows: among adults with myogenic temporomandibular disorders, which nonpharmacological therapies are most effective for reducing pain and improving maximum mouth opening? The research question was structured according to the PICO framework to ensure conceptual clarity of the population, interventions, comparators, and outcomes.

Search strategy

A comprehensive literature search was conducted in the following databases: MEDLINE (via PubMed), Embase, Cochrane Central Register of Controlled Trials (CENTRAL), Web of Science, and Google Scholar. The search was conducted from the inception of each database through to October 31, 2025, with no language restrictions. Studies published in languages other than English were considered eligible provided that a full-text English version or a reliable translation was available. The search terms included keywords related to temporomandibular joint disorders (TMD), including "myogenic temporomandibular joint disorders," "acupuncture," "exercise therapy," "extracorporeal shock wave therapy," "manual therapy," "photobiomodulation therapy," "cognitive behavioral therapy," and "occlusal splint therapy," among others. Search terms also incorporated methodological filters for randomized controlled trials and terms related to sham treatment, placebo, and usual care to ensure comprehensive identification of eligible comparator groups. A detailed search strategy, including the exact terms, date range, and methods, is available in Supplementary File 2. Additionally, reference lists of relevant articles and reviews were manually searched to identify further studies. Title/abstract and full-text screening were independently performed by LZ and ML, with any discrepancies resolved through discussion or, if necessary, arbitration by XL.

Study selection

Eligibility criteria were predefined according to the PICO framework. To be eligible for inclusion in this systematic review and network meta-analysis, studies had to meet the following criteria: (1) participants aged 18 years or older diagnosed with myogenic temporomandibular joint disorders; (2) intervention group treated with a predefined nonpharmacological modality, including acupuncture, exercise therapy, extracorporeal shock wave therapy, manual therapy, photobiomodulation therapy, cognitive behavioral therapy, occlusal splint therapy, electrotherapy, combined therapy, or central neuromodulation; (3) control group receiving either sham or placebo intervention, usual care or no-treatment control, or another predefined active nonpharmacological modality representing a different treatment node within the network, for example, manual therapy versus exercise therapy or photobiomodulation therapy versus occlusal splint therapy; (4) at least one outcome measure, including pain or maximum mouth opening, reported; (5) the study must be a published randomized controlled trial (RCT); and [6] the study must be published in a peer-reviewed journal. No language restrictions were applied; however, inclusion required availability of a full-text English version or a reliable translation to ensure accurate data extraction and methodological assessment.

Studies were excluded if they fulfilled any of the following criteria: (1) participants were children or adolescents, or diagnosed with non-myogenic temporomandibular joint disorders; (2) the intervention involved pharmacological treatments; (3) essential statistical data, such as means and standard deviations (SD), could not be extracted or estimated; (4) the full text was not accessible, or direct contact with the authors could not yield additional information. Title/abstract and full-text screening were independently performed by LZ and ML, with any discrepancies resolved through discussion or, if necessary, arbitration by XL. Inter-rater agreement before consensus resolution was calculated using percentage agreement and Cohen's kappa, and the results are reported in the Results section.

Data extraction

Following identification of eligible studies from the selected databases, references were managed using EndNote X9 citation software. LZ and ML independently extracted data from the included studies, with discrepancies resolved through consensus or consultation with XL when necessary. Extracted data included publication details (authors, title, region, year), sample size, demographic characteristics (age, sex, baseline pain score), intervention details (including intervention and control groups), compliance, and outcome measures. Detailed intervention characteristics were extracted for each treatment arm, including treatment protocol components, session duration, frequency, total treatment duration, and comparator specifications. Variability in intervention protocols, treatment duration, and follow-up schedules was also documented to inform assessment of clinical heterogeneity across studies. These data are comprehensively summarized in Supplementary Table 3.2.

For each outcome, means and standard deviations (SDs) were extracted to calculate effect sizes. When

studies reported multiple post-intervention follow-up time points, outcome data were preferentially extracted at the short-term assessment closest to the completion of the intervention period to enhance comparability across trials. If studies reported standard errors (SE) rather than SD, the SD was computed using the formula: $SD = SE \times \sqrt{n}$. In cases where neither SE nor SD were reported, standard deviation was estimated from available data, including confidence intervals (CIs), t-values, quartiles, ranges, or p-values, as outlined in *Cochrane Handbook for Systematic Reviews of Interventions* (Chap. 7.7.3). If critical data could not be obtained through these methods, authors were contacted up to four times within a six-week period to request the necessary information. LZ and ML independently extracted data from the included studies, with discrepancies resolved through consensus or consultation with XL when necessary. Inter-rater agreement for data extraction was calculated before consensus resolution using percentage agreement and Cohen's kappa.

Data coding

Interventions were categorized into the following groups based on their content: Acupuncture (ACU), Central neuromodulation (CNM), defined as non-invasive neuromodulatory interventions targeting central nervous system pathways, such as transcranial direct current stimulation,, Combined therapy (COMB), Electrotherapy (ELEC), Exercise therapy (EX), Extracorporeal shock wave therapy (ESWT), Manual therapy (MT), Occlusal splint therapy (OST), Photobiomodulation therapy (PBMT), and Cognitive behavioral therapy (CBT). Each intervention category represented a distinct and mutually exclusive treatment node defined a priori to preserve conceptual consistency within the network meta-analysis. Conventional care and placebo groups were designated as the control (CON) group. Control conditions were operationally defined to include sham interventions, placebo procedures, or usual care without additional structured nonpharmacological therapy. This predefined node structure ensured that comparisons were conducted between clearly specified therapeutic modalities rather than broadly defined nonpharmacological approaches. To support the transitivity assumption, we also reviewed the distribution of key potential effect modifiers across treatment nodes, including baseline pain severity, participant age, symptom duration where reported, intervention intensity or total treatment duration, and timing of outcome assessment. Overall, these characteristics were considered broadly comparable across nodes, although some variability and incomplete reporting remained.

Outcome measures

The primary outcomes of interest were pain and maximum mouth opening. Pain was assessed using various common pain measurement tools, including the Visual Analog Scale (VAS). The mean ± SD values before and after intervention were extracted for analysis. Maximum mouth opening was also measured using standardized methods, typically as the interincisal distance in millimeters at maximal unassisted or assisted mouth opening, measured with a ruler or caliper, as described in previous temporomandibular disorder studies [16]. To standardize outcome assessment across studies with differing follow-up schedules, the primary analysis focused on short-term effects defined as measurements obtained at or immediately after completion of the intervention. This approach was chosen to reduce clinical heterogeneity arising from variation in treatment duration and follow-up timing across trials. Longer-term follow-up data were recorded when available but were not pooled in the primary network meta-analysis due to variability in reporting intervals. Because longer-term follow-up time points were reported inconsistently and were often too sparse across treatment comparisons, no sensitivity analysis based on alternative follow-up timing was performed. For each study, detailed information regarding outcome measurement tools, evaluator identity, assessment time points, reported adverse effects, and associated statistical results was systematically extracted and is summarized in Supplementary Table 3.3 to enhance methodological transparency and reproducibility.

In cases where multiple pain scales were reported, the scale most consistently used across the included studies was selected to improve comparability and minimize heterogeneity, consistent with previous temporomandibular disorder trials that reported more than one pain assessment instrument [17]. A lower pain score indicates greater improvement, while a larger maximum mouth opening indicates better outcomes.

Risk of bias

LZ and ML independently assessed the risk of bias in the included studies using the revised Cochrane Risk of Bias Tool (RoB 2), which evaluates domains such as the randomization process, deviations from intended interventions, missing outcome data, measurement of outcomes, and selective reporting of results [18]. Any disagreements were resolved through consultation with XL. The final domain-level and overall judgments were summarized in Supplementary File 6 using a RoB 2 traffic-light plot and a summary plot, together with the detailed study-level judgments. Inter-rater agreement for the RoB 2 assessment was calculated before consensus resolution using percentage agreement and Cohen's kappa.

Data analysis

Data analysis was performed using Stata software (version 17.0; StataCorp LLC, Texas, USA). A network meta-analysis was conducted to compare the effects of different non-pharmacological treatments for myogenic temporomandibular joint disorders. Network plots were generated to illustrate the connections between various treatment comparisons, ensuring the applicability of the network structure. A random-effects model was employed to account for both within-study and between-study variability, given the anticipated clinical heterogeneity across studies. In the present review, this mainly related to differences in intervention protocols, total treatment duration, and follow-up schedules.

As the studies varied in their testing methods and tools, continuous variables were analyzed using 95% confidence intervals (CIs) and standardized mean differences (SMD). Because outcomes were measured with different instruments across studies, SMDs were used to express relative standardized effects rather than absolute changes on a single clinical scale. Heterogeneity was assessed using the I^2 statistic, with thresholds of 25%, 50%, and 75% indicating low, moderate, and high heterogeneity, respectively. The frequentist framework in Stata was employed for the network meta-analysis using the "network" and "mvmeta" packages. The relative effectiveness of treatments was ranked using the surface under the cumulative ranking curve (SUCRA), with higher SUCRA values indicating better relative efficacy. Potential publication bias was explored using comparison-adjusted funnel plots, which account for differences across treatment comparisons within the network structure. These plots were interpreted cautiously, because asymmetry may also reflect heterogeneity, inconsistency, or imbalance across comparisons rather than publication bias alone. To complement visual inspection, Egger's regression test was applied to the comparison-adjusted residuals derived from the frequentist network model, with adjustment for multi-arm trials. A p-value < 0.05 was considered suggestive of small-study effects. Given that Egger's test was originally developed for pairwise meta-analysis, its findings in the network setting were interpreted in conjunction with graphical assessment rather than as standalone evidence [19]. Additionally, prediction interval plots were generated to explore heterogeneity further and interpret the variation in effect sizes. No sensitivity or subgroup analysis based on study quality was performed, because this would have reduced network connectivity and weakened the stability of indirect comparisons. All statistical tests were two-tailed, and a p-value < 0.05 was considered statistically significant.

Results

Study selection and characteristics

A total of 6564 records were identified through the initial electronic search. After removing 3088 duplicate records, 3476 articles were screened by title and abstract. Following the exclusion of 3353 articles based on title and abstract review, 123 full-text articles were assessed for eligibility. Ultimately, 41 RCTs involving 2021 participants with myogenic TMD were included in the systematic review and network meta-analysis (Fig. 1) [10, 11, 20–58].

The studies included in this analysis were published between 2000 and 2025, with a median publication year of 2019. Of the 41 studies, 10 were from Brazil, 6 from Turkey, and 5 from Spain. The studies varied in design, with 20 single-blind RCTs, 14 double-blind RCTs, and 7 open-label RCTs. The sample sizes ranged from 11 to 99 participants, with a median of 46 participants per study. Thirty-eight studies reported the mean age of participants, which ranged from 21.7 to 46.9 years, with a median age of 32.7 years.

Of the included studies, 8 were three-arm studies, and the remaining 33 were two-arm studies. Regarding interventions, 9 studies utilized acupuncture (ACU), 10 employed photobiomodulation therapy (PBMT), 10 focused on exercise therapy (EX), 9 used manual therapy (MT), 6 combined therapies (COMB), 9 involved occlusal splint therapy (OST), 1 study utilized central neuromodulation (CNM), 4 employed electrotherapies (ELEC), 1 used extracorporeal shock wave therapy (ESWT), and 6 incorporated cognitive behavioral therapy (CBT). Twenty-four studies utilized conventional care (CON) as the control group. A detailed summary of the characteristics of the included studies is available in Supplementary File 3. Inter-rater agreement was calculated before consensus resolution for all review steps involving two independent reviewers. For study screening, the overall agreement was 95.4%, with a Cohen's κ of 0.82. For data extraction, the overall agreement was 94.8%, with a Cohen's κ of 0.89. For the RoB 2 assessment, the overall agreement was 92.6%, with a Cohen's κ of 0.86. All disagreements were resolved by discussion or consultation with a third reviewer when necessary.

Results of meta-analysis

Pain

A total of 41 studies (2021 participants with myogenic TMD) reported the outcomes for pain. Consistency analysis revealed significant heterogeneity in the results (Supplementary File 4.1), prompting further investigation using node-splitting. The comparison between ELEC and MT exhibited significant inconsistency ($p < 0.05$) (Supplementary File 5.1).

The network meta-analysis results, as shown in Fig. 2.1, display the direct comparisons and sample size

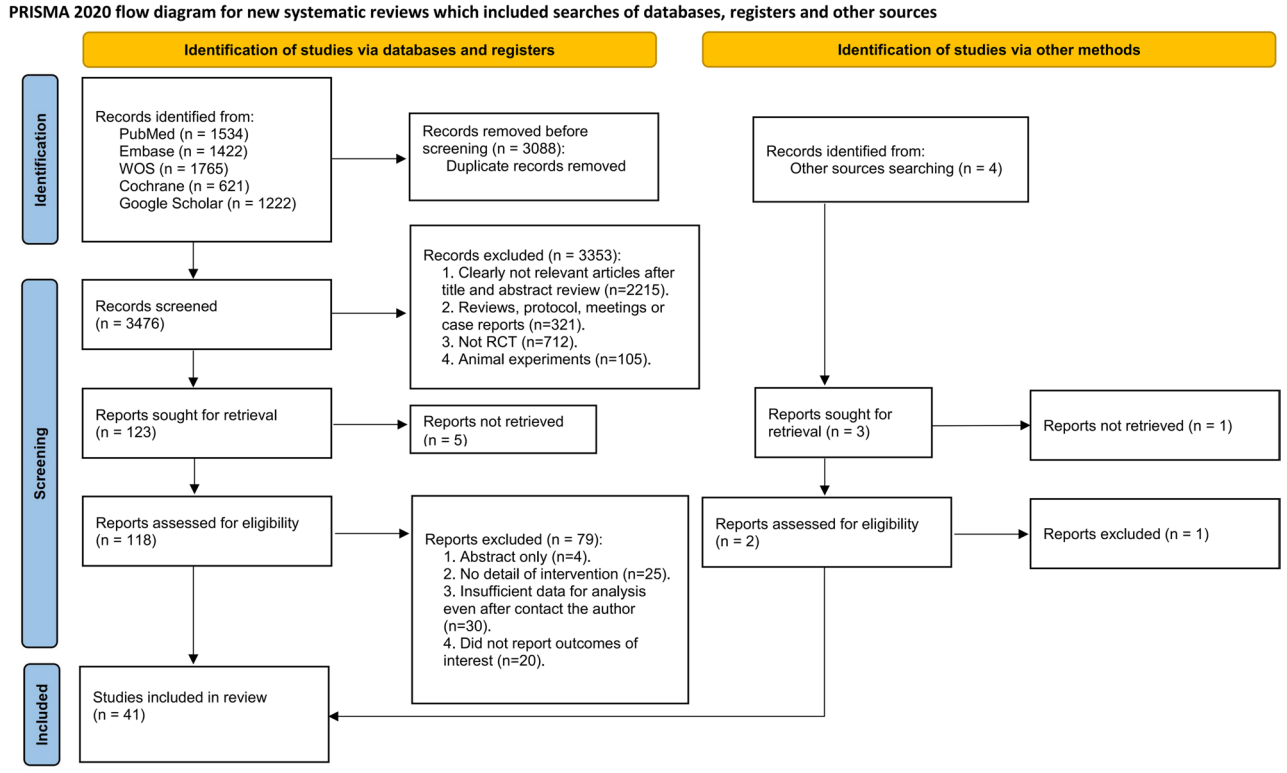


Fig. 1 PRISMA flow diagram of study selection

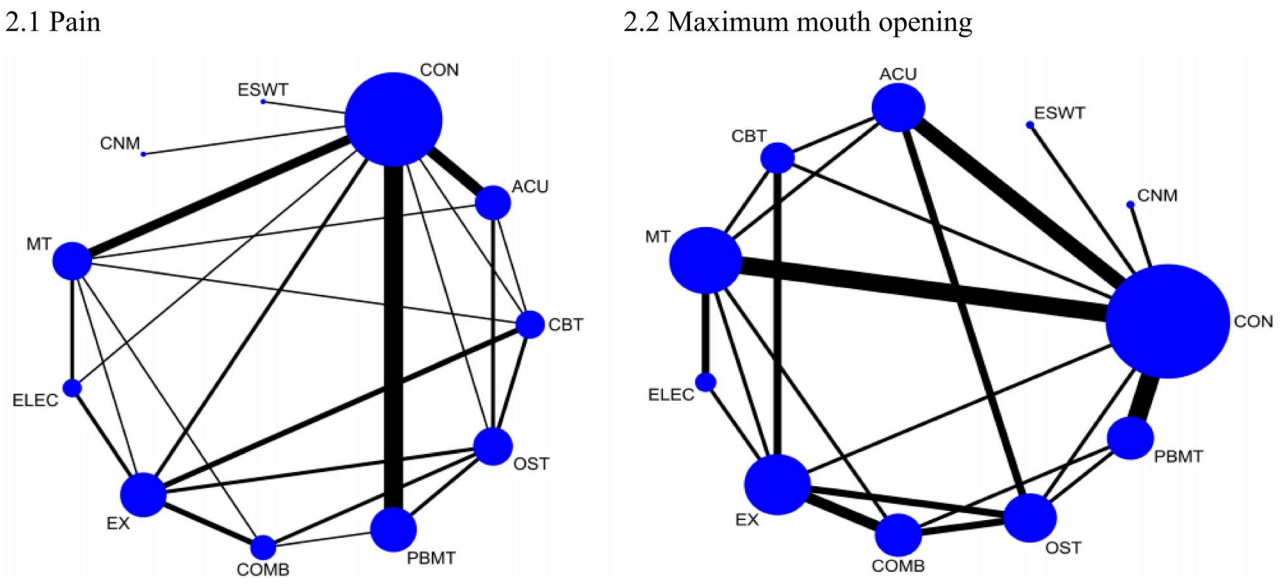


Fig. 2 Network geometry of treatment comparisons for pain and maximum mouth opening. Figure 2.1 shows the treatment network for pain, and Fig. 2.2 shows the treatment network for maximum mouth opening. Node size is proportional to the number of participants assigned to each treatment, and line thickness is proportional to the number of direct comparisons between treatment pairs. ACU, acupuncture; CBT, cognitive behavioral therapy; CNM, central neuromodulation; COMB, combined therapy; CON, conventional care, placebo, or no-treatment control; ELEC, electrotherapy; ESWT, extracorporeal shock wave therapy; EX, exercise therapy; MT, manual therapy; OST, occlusal splint therapy; PBMT, photobiomodulation therapy

distribution across different non-pharmacological treatments. The pain network was connected, with conventional care acting as the main common comparator. Although several contrasts were informed by direct evidence, some active treatment comparisons relied partly on indirect evidence through shared comparators. According to the Surface Under the Cumulative Ranking Curve (SUCRA) rankings (Fig. 3.1), the top three

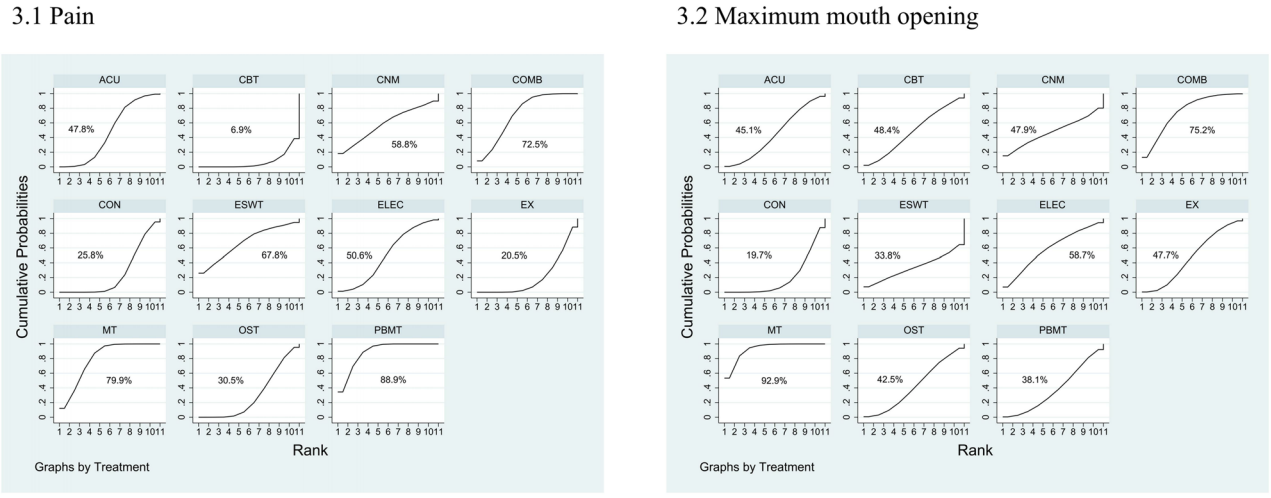


Fig. 3 SUCRA plots for treatment ranking in pain and maximum mouth opening. Figure 3.1 presents the cumulative ranking probabilities for pain, and Fig. 3.2 presents the cumulative ranking probabilities for maximum mouth opening. Higher SUCRA values indicate a greater probability that the treatment ranks among the most effective options for the corresponding outcome. SUCRA, surface under the cumulative ranking curve; ACU, acupuncture; CBT, cognitive behavioral therapy; CNM, central neuromodulation; COMB, combined therapy; CON, conventional care, placebo, or no-treatment control; ELEC, electrotherapy; ESWT, extracorporeal shock wave therapy; EX, exercise therapy; MT, manual therapy; OST, occlusal splint therapy; PBMT, photobiomodulation therapy

Table 1 League table of network meta-analysis results for pain and maximum mouth opening. League table of network meta-analysis results for pain

PBMT										
-0.27 (-1.14,0.61)	MT									
-0.45 (-1.48,0.57)	-0.19 (-1.20,0.83)	COMB								
-0.43 (-2.42,1.56)	-0.17 (-2.19,1.86)	0.02 (-2.11,2.15)	ESWT							
-0.68 (-2.78,1.41)	-0.42 (-2.55,1.71)	-0.23 (-2.46,2.00)	-0.25 (-3.02,2.52)	CNM						
-0.98 (-2.13,0.17)	-0.71 (-1.74,0.31)	-0.53 (-1.76,0.71)	-0.55 (-2.71,1.62)	-0.29 (-2.56,1.97)	ELEC					
-1.07 (-1.93,-0.21)	-0.81 (-1.70,0.09)	-0.62 (-1.70,0.46)	-0.64 (-2.66,1.38)	-0.39 (-2.51,1.74)	-0.09 (-1.28,1.09)	ACU				
-1.43 (-2.29,-0.56)	**-1.16 (-2.10,-0.22)**	**-0.97 (-1.91,-0.04)**	-0.99 (-3.05,1.07)	-0.74 (-2.90,1.42)	-0.45 (-1.62,0.73)	-0.35 (-1.23,0.52)	OST			
-1.49 (-2.07,-0.91)	**-1.23 (-1.92,-0.54)**	**-1.04 (-2.00,-0.09)**	-1.06 (-2.96,0.84)	-0.81 (-2.82,1.21)	-0.51 (-1.54,0.52)	-0.42 (-1.10,0.26)	-0.07 (-0.85,0.72)	CON		
-1.64 (-2.55,-0.73)	**-1.37 (-2.26,-0.49)**	**-1.19 (-2.05,-0.32)**	-1.21 (-3.26,0.85)	-0.95 (-3.11,1.21)	-0.66 (-1.70,0.38)	-0.57 (-1.51,0.38)	-0.21 (-1.03,0.61)	-0.15 (-0.93,0.64)	EX	
-2.00 (-2.99,-1.00)	**-1.73 (-2.71,-0.75)**	**-1.55 (-2.61,-0.48)**	-1.57 (-3.66,0.53)	-1.31 (-3.51,0.88)	-1.02 (-2.22,0.18)	-0.93 (-1.91,0.06)	-0.57 (-1.46,0.31)	-0.51 (-1.38,0.37)	-0.36 (-1.18,0.46)	CBT

Note: Bold values indicate statistically significant differences

interventions for reducing pain were PBMT (88.9%), MT (79.9%), and COMB (72.5%), with CBT ranking the lowest (6.9%). These rankings should be interpreted alongside the effect estimates, because SUCRA reflects relative ranking probability rather than certainty of superiority, and rankings may be unstable where direct evidence is sparse. Furthermore, as presented in Table 1, all comparative estimates are reported as SMDs with 95% confidence intervals. PBMT significantly reduced pain compared with CBT – 2.00 (-2.99, -1.00), EX -1.64 (-2.55, -0.73), CON – 1.49 (-2.07, -0.91), OST – 1.43 (-2.29, -0.56), and ACU – 1.07 (-1.93, -0.21). Similarly, MT was more effective than CBT – 1.73 (-2.71, -0.75), EX -1.37 (-2.26, -0.49), CON – 1.23 (-1.92, -0.54), and OST – 1.16 (-2.10, -0.22). COMB was also superior to CBT – 1.55 (-2.61, -0.48), EX -1.19 (-2.05, -0.32), CON – 1.04 (-2.00, -0.09), and OST – 0.97 (-1.91, -0.04) for pain reduction.

Maximum mouth opening

Thirty-one studies (1463 participants) assessed the effect of various non-pharmacological treatments on maximum mouth opening. Consistency analysis revealed

Table 2 League table of network meta-analysis results for maximum mouth opening

MT										
0.89 (-1.13,2.91)	COMB									
1.42 (-0.85,3.70)	0.54 (-2.26,3.33)	ELEC								
1.87 (-0.31,4.05)	0.98 (-1.45,3.41)	0.44 (-2.50,3.38)	CBT							
1.97 (-2.15,6.09)	1.08 (-3.26,5.42)	0.55 (-4.08,5.17)	0.10 (-4.28,4.49)	CNM						
1.85 (0.05,3.64)	0.96 (-0.76,2.68)	0.42 (-2.08,2.92)	-0.02 (-2.05,2.01)	-0.12 (-4.37,4.12)	EX					
2.01 (0.12,3.90)	1.12 (-1.11,3.35)	0.58 (-2.22,3.39)	0.14 (-2.15,2.43)	0.04 (-4.11,4.18)	0.16 (-1.89,2.21)	ACU				
2.05 (0.00,4.10)	1.16 (-0.79,3.11)	0.62 (-2.21,3.46)	0.18 (-2.23,2.59)	0.08 (-4.19,4.35)	0.20 (-1.66,2.06)	0.04 (-1.87,1.95)	OST			
2.23 (0.19,4.28)	1.34 (-0.93,3.62)	0.81 (-2.11,3.72)	0.37 (-2.15,2.88)	0.26 (-3.89,4.41)	0.39 (-1.81,2.59)	0.22 (-1.88,2.32)	0.18 (-1.99,2.36)	PBMT		
2.67 (-1.39,6.73)	1.78 (-2.50,6.07)	1.25 (-3.33,5.82)	0.81 (-3.52,5.14)	0.70 (-4.70,6.10)	0.83 (-3.36,5.02)	0.66 (-3.42,4.75)	0.62 (-3.59,4.84)	0.44 (-3.65,4.53)	ESWT	
2.79 (1.33,4.26)	1.91 (-0.10,3.91)	1.37 (-1.20,3.94)	0.93 (-1.17,3.03)	0.82 (-3.03,4.67)	0.95 (-0.85,2.74)	0.79 (-0.76,2.33)	0.74 (-1.10,2.59)	0.56 (-1.00,2.12)	0.12 (-3.67,3.91)	CON

Note: Bold values indicate statistically significant differences

PBMT Photobiomodulation therapy, *MT* Manual therapy, *COMB* Combination therapy, *ESWT* Extracorporeal shock wave therapy, *CNM* Central neuromodulation, *ELEC* Electrotherapy, *ACU* Acupuncture, *OST* Occlusal splint therapy, *CON* Control (usual care or placebo), *EX* Exercise therapy, *CBT* Cognitive behavioral therapy

no significant overall inconsistency for this outcome ($p = 0.981$) (Supplementary File 4.2), and further node-splitting analysis confirmed the absence of significant inconsistency ($p > 0.05$) across all comparisons (Supplementary File 5.2).

The network meta-analysis results, presented in Fig. 2.2, show the direct comparisons and sample size distribution for maximum mouth opening outcomes. This network was likewise connected, but several active treatment contrasts were informed by limited direct evidence, increasing the relative contribution of indirect comparisons. According to the SUCRA rankings (Fig. 3.2), the top three interventions for improving maximum mouth opening were MT (92.9%), COMB (75.2%), and ELEC (58.7%), with the CON group showing the lowest efficacy (19.7%). These ranking results should be interpreted cautiously and in conjunction with the underlying comparative estimates. As shown in Table 2, MT was significantly more effective than CON 2.79 (1.33, 4.26), PBMT 2.23 (0.19, 4.28), ACU 2.01 (0.12, 3.90), and EX 1.85 (0.05, 3.64) in improving maximum mouth opening.

Risk of bias and publication bias

Of the 41 RCTs included in the analysis, 22 were assessed as having a low risk of bias, while 19 studies were deemed to have some concerns. In terms of randomization, 36 studies had a low risk of bias, with 5 studies having some concerns. Regarding deviations from intended interventions, 34 studies were rated as having a low risk of bias, and 7 studies had some concerns. With respect to missing outcome data, 29 studies were categorized as having a low risk of bias, and 12 studies had some concerns. For outcome measurement, 39 studies had a low risk of bias, and 2 studies had some concerns. All 41 studies were rated as having a low risk for selective reporting bias. The complete RoB 2 assessment is presented in Supplementary File 6, including the study-level traffic-light plot, the domain-level summary plot, and the detailed judgments for each included trial.

Potential publication bias was assessed using funnel plots (Supplementary File 7). The scatterplots around the vertical axis showed varying degrees of asymmetry, suggesting the possibility of publication bias. Specifically, the funnel plots for maximum mouth opening (S5.1-S5.2) displayed some degree of asymmetry. The Egger's test for maximum mouth opening yielded a p-value < 0.05 (S5.2), indicating the potential presence of publication bias, which warrants caution in interpreting the results. This finding also reduces confidence in the relative ranking of interventions for maximum mouth opening, particularly the high ranking of MT, because small-study effects may have exaggerated the apparent comparative advantage.

Discussion

This systematic review and NMA included 41 RCTs involving 2021 adults with myogenic TMD and systematically compared nonpharmacological interventions for pain and maximum mouth opening. Three main findings were identified. First, for pain, PBMT achieved the highest overall ranking, suggesting the most favorable short-term comparative profile among the interventions assessed. Second, although MT and COMB ranked second and third for pain, respectively, both showed statistically significant analgesic benefits in network comparisons versus several commonly used strategies, supporting a consistent and clinically meaningful effect on

pain outcomes. Third, for maximum mouth opening, MT ranked highest in the network and was the only treatment demonstrating clear superiority over multiple key comparators within the network, suggesting a potential advantage for short-term functional improvement. However, confidence in this ranking is moderated by evidence of possible publication bias for the MMO outcome. These findings should, however, be interpreted with caution in view of heterogeneity across studies, some concerns regarding risk of bias, and the predominance of short-term outcome assessment. These results offer a comparative, outcome-oriented evidence base for selecting nonpharmacological therapies in adult myogenic TMD and underscore that the interventions most effective for pain relief may not be identical to those that optimize mandibular function.

Pain intensity is the clinical outcome that most directly determines whether adults with myogenic TMD seek care, adhere to conservative programs, and perceive treatment as successful, even when functional limitation is modest. In the present random effects network, PBMT was positioned as the most favorable nonpharmacological option for pain relief, with a higher overall ranking than other commonly used conservative strategies and with network estimates indicating benefit over several key comparators (including CON and multiple active modalities). This pattern is broadly concordant with prior randomized trials and conventional pairwise syntheses evaluating low level laser and related photo biomodulation approaches in TMD, which have generally suggested analgesic effects but have also reported variable magnitude and certainty, largely attributable to differences in patient phenotyping, laser parameters, and treatment dosing [59]. The current analysis extends that literature by integrating indirect evidence across a connected network and by focusing specifically on adult myogenic TMD, a subgroup in which muscle based nociceptive mechanisms are expected to be particularly responsive to local biophysical modulation [60]. Photo biomodulation is thought to influence mitochondrial photo acceptors, enhance cellular energy availability, and improve local microcirculation, while also modulating inflammatory signaling and oxidative stress within sensitized masticatory muscles. These effects could attenuate peripheral nociceptor activation at myofascial trigger points and reduce muscle tenderness, thereby lowering pain ratings even in the absence of large changes in mandibular range of motion [61, 62]. At the same time, the PBMT protocols across trials varied substantially in wavelength, energy density, treatment sites, and session frequency, and sham procedures were generally feasible, which may have strengthened internal validity but also contributed to between study variability [63]. Accordingly, the observed ranking should be interpreted as comparative evidence of relative analgesic performance rather than as definitive proof of superiority for any single PBMT regimen. In addition, because the present analysis was based primarily on short-term outcomes, the durability of benefit remains uncertain. Some estimates were large in standardized units, which may partly reflect differences in outcome scaling, within-study variability, and the contribution of small or heterogeneous comparisons. These values should therefore be interpreted cautiously in clinical terms.

Building on the finding that PBMT ranked highest for pain, the network also identified MT and COMB as comparably strong analgesic options. Although their overall rankings were second and third, respectively, both interventions showed statistically significant pain reductions versus several commonly used approaches in the network, including CON and selected active comparators. This pattern suggests that their effects on pain were not limited to favorable placement in rankings but reflected a consistent signal across the available direct and indirect evidence. These findings are generally consistent with prior trials and pairwise syntheses indicating that MT can improve short-term pain in myogenic TMD and that multimodal programs often outperform single-component care [64]. Variation in reported effects across the literature is expected, given the breadth of techniques classified as MT, differences in treatment dose and targets, and heterogeneity in multimodal protocols and co-interventions [65]. MT may reduce pain by decreasing myofascial trigger point sensitivity, improving soft-tissue mobility and local perfusion, and modulating afferent input from the masticatory and cervical regions, with additional contributions from neurophysiologic pain modulation [36, 66]. COMB may enhance analgesia by addressing multiple drivers simultaneously, such as peripheral muscle tenderness, motor control dysfunction, and behavioral contributors, thereby providing a broader therapeutic coverage than any single modality [67]. Nonetheless, because the content and dosing of COMB varied across trials, the present results support multimodal care as a concept rather than any specific combination regimen. In real-world care, MT may be more immediately applicable than device-based therapies in some settings, but its effectiveness is likely to depend on therapist training, manual skill, and familiarity with temporomandibular and cervical musculoskeletal assessment. COMB may be clinically attractive for patients with mixed pain and functional impairment, although it may also require greater treatment time, coordination, and health care resources.

In contrast to PBMT, MT, and COMB, CBT and EX occupied the lowest positions for pain and, in this network, appeared no more favorable than CON. This finding does not negate prior reports suggesting that

behavioral interventions and exercise-based rehabilitation can be useful in TMD, but it is consistent with the observation that these approaches often yield smaller effects on pain intensity than on related domains such as pain interference, coping, self-efficacy, and functional performance [9, 12]. In several included trials, CBT was delivered in brief formats that emphasized education, relaxation, or self-regulation training rather than a full course of structured cognitive and behavioral skill acquisition, and it was sometimes compared with active usual care packages that already included reassurance and self-management guidance, which may have reduced the detectable incremental analgesic effect. For EX, protocols were heterogeneous and frequently relied on unsupervised practice, with incomplete adherence reporting; analgesic benefits may depend on adequate dose, correct technique, and longer follow up than was available in many studies, and early stretching or loading can transiently aggravate symptoms in sensitized masticatory muscles. These considerations suggest that the low ranking for CBT and EX most likely reflects limited and variable evidence for short term reductions in pain scores rather than evidence that these interventions lack clinical value when broader patient centered outcomes are considered. At the same time, exercise-based and behavioral strategies may remain relevant in routine practice because they are generally more accessible, less equipment-dependent, and easier to incorporate into long-term self-management pathways.

Maximum mouth opening is a clinically salient functional endpoint in myogenic TMD because it reflects the capacity to chew, speak, and perform daily activities without provoking guarding or pain-related avoidance. In the present network, MT emerged as the most effective intervention for this outcome and was the only treatment that showed clear superiority over multiple key comparators in the network, including CON and several active modalities. This pattern is broadly consistent with prior trials and pairwise syntheses suggesting that hands on approaches can improve mandibular mobility, although the magnitude of benefit has varied across studies owing to differences in technique, treatment targets, and baseline restriction severity [66]. A plausible explanation is that MT directly addresses biomechanical and neuromuscular contributors to limited opening. Soft-tissue mobilization and myofascial techniques may reduce muscle hypertonicity and trigger point irritability, improve tissue compliance, and permit greater excursion of the mandible, while joint oriented maneuvers and cervical interventions may normalize coupled motion and reduce protective co-contraction [68]. Neurophysiologic effects, including reduced nociceptive input and altered motor output, may further diminish guarding and facilitate movement. The observation that MT ranked highly for both pain and maximum mouth opening is clinically coherent, as reductions in peripheral muscle tenderness and central excitability can lower pain while simultaneously enabling fuller, less guarded mandibular motion [69]. Nonetheless, because MT protocols varied across trials and follow up was often short, these findings should be interpreted as evidence supporting MT as a leading functional strategy rather than as a definitive estimate for any single technique or dose. In addition, the asymmetry observed in the funnel plot and the significant Egger's test for MMO suggest possible small-study effects or publication bias. This means that the magnitude of benefit for MT may be overestimated and that its top ranking for MMO should be interpreted with reduced confidence. Clinically, this may make MT particularly relevant when restoration of mandibular mobility is the primary treatment goal, provided that appropriately trained clinicians are available.

The present findings have several implications for clinical decision making in adult myogenic TMD. PBMT may be considered when short-term pain relief is the main treatment objective, whereas MT may be considered when improvement in maximum mouth opening is of primary concern. MT and COMB may be particularly relevant for patients with concurrent pain and functional limitation, although their implementation may depend on therapist expertise, treatment time, and local service capacity. By contrast, exercise-based and behavioral approaches may be more accessible and easier to integrate into long-term self-management, despite their less favorable short-term ranking for pain in the present analysis. Accordingly, treatment selection should be guided not only by comparative efficacy, but also by patient preference, resource availability, feasibility of delivery, and the level of clinical expertise required for implementation.

This study has several strengths. First, it synthesized the totality of randomized evidence for adult myogenic TMD using a frequentist random-effects network meta-analysis, allowing simultaneous comparison and ranking of multiple nonpharmacological options even when head to head trials were sparse. Second, eligibility criteria and intervention coding were restricted to myogenic presentations and prespecified treatment nodes, improving clinical interpretability and reducing dilution from mixed or nonmuscular TMD phenotypes. Several limitations should be considered. First, clinical and methodological heterogeneity was substantial, particularly for pain, as trials differed in diagnostic criteria, outcome instruments, and follow up duration. Intervention protocols also varied across studies in dose, frequency, duration, and technical delivery, which may have contributed to heterogeneity. Second, the primary analysis was restricted to short-term outcomes measured at or near completion of

treatment, so the findings do not establish durability of benefit. A significant inconsistency signal was detected for at least one comparison, which may affect the stability of some indirect estimates. Third, although key potential effect modifiers were reviewed during network construction, transitivity was not formally assessed, which may limit confidence in some indirect comparisons. Fourth, we did not perform a formal certainty assessment, such as GRADE or CINeMA, and therefore the certainty of the network estimates could not be formally established. Fifth, the evidence base was uneven across nodes: PBMT, MT, and EX were informed by more trials than CNM or ESWT, and many comparisons relied on small, single center studies with limited sample sizes, which increases imprecision and may inflate apparent differences in ranking. This may also have contributed to the large effect estimates observed in some comparisons. Sixth, reporting of adherence, cointerventions, and blinding was variable, and open label designs were present. Although 22 studies were judged to be at low risk of bias overall, the remaining studies raised some concerns, and this should be taken into account when interpreting the comparative estimates. Although comparison-adjusted funnel plots were used to explore small-study effects within the network structure, formal regression-based tests such as Egger's test were interpreted cautiously, as their application in network meta-analysis may be influenced by multi-arm correlations and between-comparison heterogeneity. Together with the asymmetry observed for maximum mouth opening, these factors raise the possibility of performance and publication biases that warrant cautious interpretation of comparative effects. Overall, these considerations suggest that the present findings are most appropriately interpreted as evidence of relative short-term comparative effects rather than definitive proof of treatment superiority. Because risk-of-bias judgments were unevenly distributed across treatment comparisons, no sensitivity or subgroup analysis based on study quality was performed.

Conclusion

In this random-effects network meta-analysis of 41 randomized trials (2021 adults) with myogenic temporomandibular disorders, PBMT showed the most favorable short-term comparative profile for pain reduction, whereas MT ranked highest for improving maximum mouth opening. MT and COMB also showed favorable analgesic effects within the network, while CBT and EX showed limited short-term effects on pain intensity. However, these findings should be interpreted cautiously in light of heterogeneity across studies, variability in intervention protocols, and the predominance of short-term outcomes. PBMT and MT may be considered promising options when pain relief or improvement in mandibular mobility is the primary treatment goal, but high-quality head-to-head trials are still needed to confirm these comparative findings.

Supplementary Information

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Supplementary Material 1.

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Authors' contributions

Lei Zhang: Data curation, Formal Analysis, Methodology, Software, Writing – original draft. Minghe Liu: Data curation, Software, Writing – original draft. Xiaojing Li: Formal Analysis, Methodology, Writing – original draft. Chanyuan Zhao: Conceptualization, Supervision, Validation, Visualization, Writing – review & editing. All authors contributed to the manuscript and approved the final version for submission.

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Data availability

All data generated or analysed during this study are included in this published article and its supplementary information files.

Declarations

Ethics approval and consent to participate

This is a systematic review and meta-analysis, ethics approval and consent to participate are not applicable.

Consent for publication

Not applicable. The manuscript does not include the participant's identification image or other personal or clinical details.

Competing interests

The authors declare no competing interests.

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